Three two-cycles in a strictly convex monotone recurrence

Abstract

We give an explicit real-analytic, nonnegative, coordinatewise increasing and jointly strictly convex function whose second-order scalar recurrence has exactly three prime two-cycles. These give six phase-distinct nonconstant two-periodic solutions, in the class considered by Berger and Duh. Two cycles are ambient saddles that attract along the boundary of the lower equilibrium's basin; the third cycle repels there. The global basin geometry is supplied by the existing convex-monotone recurrence theorem. We certify the exact count and the stability at the actual analytic states using a finite rational cell calculation and uniform softening estimates. The construction is a concrete realization for a question highlighted in the earlier work; interpolation, smoothing and the general multi-cycle stability mechanisms are prior tools.

Keywords: second-order difference equation; convex monotone map; two-cycle; basin boundary; analytic realization

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1 The recurrence and the example

Write $I = [0, \infty)$. Consider the autonomous recurrence and its two-position state map

$$x_{n+1} = g(x_n, x_{n-1}), \qquad n \geq 0, \qquad T(u, v) = (v, g(v, u)), \tag{1}$$

with arbitrary $(x_{-1}, x_0) \in I^2$. Berger and Duh [1] study the class

$$\mathcal{G} = \{g : I^2 \to I : \ g \in C^1, \ g \text{ is jointly strictly convex}, \ \nabla g \geq 0\}. \tag{2}$$

Their Theorem 2.2 describes a bounded convex basin and its boundary dynamics. In Remark 4.6 they discuss the possible number of nonconstant two-periodic solutions and record explicit examples with four such solutions. Their counting convention distinguishes $(u, v, u, v, \ldots)$ from its shift $(v, u, v, u, \ldots)$. Those two sequences form one prime two-cycle.

We give a fixed analytic function in (2) with six such sequences, or three prime two-cycles. The point of the example is global admissibility together with a complete count. Let $B_\infty(c, r)$ denote the closed maximum-norm box of radius r centered at c, and put

$$r = \frac{1}{100}, \qquad \zeta = \frac{173592647}{49408820}, \qquad c_1 = (1, 19/4), \quad c_2 = (2, 9/2), \quad c_3 = (3, 4). \tag{3}$$

Theorem 1. *The explicit function g in (8) is real analytic on $\mathbb{R}^2$, strictly positive, has positive first partial derivatives, and has a positive-definite Hessian at every point. In particular its restriction belongs to $\mathcal{G}$.*

The state map (1) *has exactly two equilibria and exactly three prime two-cycles in* I^2. *The equilibria are* (ξ, ξ), *where* $0 < \xi < r$, *and* $(\eta, \eta) \in B_\infty((\zeta, \zeta), r)$. *For each* $i = 1, 2, 3$ *there is exactly one two-periodic state* z_i *in* $B_\infty(c_i, r)$; *its partner is its coordinate swap. These eight states are all the solutions of* $T^2 z = z$.

Every state in $[0, r]^2$ *converges to the lower equilibrium, which is a local sink. At each of the other seven states,* DT^2 *has two distinct positive eigenvalues* $\lambda_+ > \lambda_-$ *with* $\lambda_+ > 1$. *The cycles through* z_1, z_3 *have* $\lambda_- < 1$ *and are ambient hyperbolic saddles. The cycle through* z_2 *has* $\lambda_- > 1$ *and is an ambient repeller. Both eigenvalues of* $DT(\eta, \eta)$ *have modulus greater than one.*

Let A_ξ *be the basin of the lower equilibrium and let* A_∞ *be the set of states whose scalar iterates tend to* $+\infty$. *The prior basin theorem applies and gives*

$$\mathrm{cl}(A_\xi) \cup A_\infty = I^2, \tag{4}$$

where A_ξ *is nonempty, relatively open, bounded and convex, and contains* $(0, 0)$. *Every trajectory on its relative boundary has convergent odd and even subsequences. The three cycles and the upper equilibrium lie on this boundary. The cycles through* z_1, z_3 *locally attract within the boundary; the middle cycle and the upper equilibrium locally repel within it.*

The periodic states of the analytic map are localized near the rational centers in (3); they are not asserted to equal those centers. Conditional attraction means attraction restricted to the invariant boundary, not an open ambient basin. The positive lower equilibrium is excluded from every assertion about expanding return multipliers. A constant trajectory is not one of the six nonconstant phase-distinct solutions.

Several-cycle dynamics in monotone recurrences is established prior work. Bilgin, Kulenović and Pilav [2], and Bertrand and Kulenović [3], treat three prime two-cycles in sigmoid and Beverton–Holt models. Those bounded models are outside the global convex class (2). The polynomial analysis of Bektešević, Kulenović and Pilav [4] also supplies conditional multi-cycle basin descriptions. We use the global convex-basin theorem of [1]; we do not introduce this geometry or its stability mechanism.

Convex monotone interpolation and analytic convex approximation are standard tools; see [5, Section 6.5.5] and [6]. Here finite positive softplus sums give the required global properties directly. The remaining work is to exclude every extra analytic root and transfer the return type to the actual softened states. Our literature comparison is bounded. The general-polynomial paper [7] was accessible only in part, so complete coverage of its statements has not been established. No firstness or arbitrary-multiplicity claim is made.

2 One fixed analytic function

Set

$$\sigma = 10^{-12}, \qquad S_\sigma(t) = \sigma \log(1 + e^{t/\sigma}) \quad (t \in \mathbb{R}). \tag{5}$$

The thirteen knots, indexed from zero, are

$$\begin{aligned}(k_0, \dots, k_{12}) = \Big(0, \frac{19}{20}, \frac{21}{20}, \frac{39}{20}, \frac{41}{20}, \frac{59}{20}, \frac{61}{20},\\ \frac{79}{20}, \frac{81}{20}, \frac{89}{20}, \frac{91}{20}, \frac{47}{10}, \frac{24}{5}\Big).\end{aligned} \tag{6}$$

Every a_i, b_i equals $1/1000$, except for the following six replacements:

$$\begin{aligned} a_1 &= \frac{147619}{1188375}, \quad a_5 = \frac{1176727}{9507000}, \quad a_7 = \frac{2618587}{9507000},\\ a_{11} &= \frac{2120763}{1584500}, \quad b_2 = \frac{5992813}{4753500}, \quad b_{11} = \frac{3713449}{3169000}. \end{aligned} \tag{7}$$

Define

$$P(t) = \sum_{i=0}^{12} a_i S_\sigma(t - k_i), \qquad Q(t) = \sum_{i=0}^{12} b_i S_\sigma(t - k_i), \qquad g(u, v) = P(u) + Q(v). \tag{8}$$

All constants are fixed; none depends on a horizon or a choice of initial state. Exact summation gives

$$\sum_i a_i = \frac{17786407}{9507000} < 2, \qquad \sum_i b_i = \frac{464611}{190140} < 3. \tag{9}$$

Lemma 1. *The function g has all the global regularity, range, monotonicity and convexity properties stated in Theorem 1. If $H(t) = \max\{t, 0\}$, then*

$$0 < S_\sigma(t) - H(t) \le \sigma \log 2 < \sigma \qquad (t \in \mathbb{R}). \tag{10}$$

For $|t| \ge d = 1/25$,

$$|S'_\sigma(t) - H'(t)| \le e^{-d/\sigma} \le 2(\sigma/d)^2 = \frac{1}{8 \cdot 10^{20}} < \sigma. \tag{11}$$

Proof. The logarithm in (5) has a positive real-analytic argument. Thus S_σ is real analytic at every real point. Its value is positive, and

$$S'_\sigma(t) = \frac{1}{1 + e^{-t/\sigma}} > 0, \qquad S''_\sigma(t) = \frac{e^{-t/\sigma}}{\sigma(1 + e^{-t/\sigma})^2} > 0. \tag{12}$$

Positive coefficients in (8) give positive values and first derivatives. The Hessian of g is $\operatorname{diag}(P''(u), Q''(v)) \succ 0$ everywhere. Integration along any nonconstant segment proves joint strict convexity, including along either coordinate axis. There is no claim of a uniform strong-convexity modulus.

The identity $S_\sigma(t)-H(t)=\sigma\log(1+e^{-|t|/\sigma})$ proves (10). Away from zero the derivative error is at most $e^{-|t|/\sigma}$. Finally $e^x \geq x^2/2$ for $x>0$ gives the rational bound in (11). ∎

The literal state convention in (1) gives

$$T^2(u,v)=(u,v) \quad \Longleftrightarrow \quad F(u,v):=\begin{pmatrix} P(u)+Q(v)-v \\ P(v)+Q(u)-u \end{pmatrix}=0. \tag{13}$$

At any such root, $T(u,v)=(v,u)$. It therefore suffices to count all zeros of F on the full quadrant, and then determine which are diagonal.

3 An exact finite root certificate

Replace S_σ by H in (8), and denote the resulting functions by P_0, Q_0. Let F_0 be the corresponding residual in (13). Equations (9) and (10) give the uniform estimate

$$\|F-F_0\|_\infty < \varepsilon, \qquad \varepsilon = 5\sigma, \tag{14}$$

on all of $\mathbb{R}^2$. This bound will permit an exclusion argument on unbounded cells.

For $0 \leq i < 12$, put $J_i=[k_i,k_{i+1}]$, and let $J_{12}=[k_{12},\infty)$. These thirteen closed intervals cover I. Define rational numbers

$$A_i=\sum_{h\leq i} a_h, \quad B_i=\sum_{h\leq i} b_h, \quad C_i=\sum_{h\leq i} a_h k_h, \quad D_i=\sum_{h\leq i} b_h k_h. \tag{15}$$

On J_i, including its endpoints, $P_0(t)=A_it-C_i$ and $Q_0(t)=B_it-D_i$. Hence on $J_i\times J_j$,

$$F_0(z)=M_{ij}z-b_{ij}, \qquad M_{ij}=\begin{pmatrix} A_i & B_j-1 \\ B_i-1 & A_j \end{pmatrix}, \quad b_{ij}=\begin{pmatrix} C_i+D_j \\ D_i+C_j \end{pmatrix}. \tag{16}$$

The adjacent formulas agree on common edges. With $\Delta_{ij}=\det M_{ij}$, define

$$z_{ij}=M_{ij}^{-1}b_{ij}, \qquad K_{ij}=\|M_{ij}^{-1}\|_\infty=\frac{\max\{A_j+|B_j-1|, A_i+|B_i-1|\}}{|\Delta_{ij}|}, \tag{17}$$

whenever the determinant is nonzero. Let

$$\delta_{ij}=\operatorname{dist}_\infty(z_{ij}, J_i\times J_j). \tag{18}$$

For a bounded interval $[l,u]$, the scalar distance of t to it is $\max\{l-t,t-u,0\}$; for $[l,\infty)$ it is $\max\{l-t,0\}$. Thus every quantity in (17)–(18) is rational.

Lemma 2. *All* 169 *determinants* Δ_{ij} *are nonzero. Precisely eight cells have* $\delta_{ij}=0$*, with centers listed in Table 1. The exact bounds are*

$$\min_{i,j}|\Delta_{ij}|=\frac{28141190147}{9038304900000}>0, \tag{19}$$

$$\max_{i,j}K_{ij}=\frac{7195825483200}{28141190147}<256, \tag{20}$$

$$\min_{\delta_{ij}>0}\frac{\delta_{ij}}{K_{ij}}=\frac{6804206}{29929224375}>\frac{1}{5000}. \tag{21}$$

For each of the seven nonzero centers in the table, its closed r*-box lies in the open quadrant, in the interior of its cell, and at coordinate distance at least* $d=1/25$ *from every knot.*

Cell (i,j)	z_{ij}	Cell (i,j)	z_{ij}
$(0,0)$	$(0,0)$	$(6,6)$	(ζ,ζ)
$(1,11)$	$(1,19/4)$	$(11,1)$	$(19/4,1)$
$(3,9)$	$(2,9/2)$	$(9,3)$	$(9/2,2)$
$(5,7)$	$(3,4)$	$(7,5)$	$(4,3)$

Table 1: All feasible affine roots of F_0. They localize the analytic roots of F but are not asserted to equal them.

Proof. This is a finite rational calculation from (6)–(7) and (15)–(18). For clarity, it can be reproduced without a root solver. For each $i,j \in \{0,\dots,12\}$, calculate M_{ij} and b_{ij}, verify $\Delta_{ij} \neq 0$, invert the two-by-two matrix, and evaluate the displayed distance formulas. The zero-distance cells give exactly Table 1. Taking the three rational extrema gives (19)–(21). The maximum in (20) occurs at $(4,7)$ and $(7,4)$; the minimum in (21) occurs at $(3,10)$ and $(10,3)$.

The companion file `certificate.json` records every matrix, root, inverse norm and distance for all 169 cells. The supplied `verify_certificate.py` regenerates these rational formulas using integer fractions, verifies the extrema, and compares every entry. It also substitutes the eight roots into the original hinge sums. No floating-point comparisons or numerical root lists enter these assertions.

The final box statement follows by comparing the rational centers with the knot endpoints. Each off-diagonal center is at distance $1/20$ from its nearest knot, and direct rational comparison gives a larger distance for ζ. Subtracting $r = 1/100$ leaves $1/25$. ∎

Proposition 3.1. Every zero of F in I^2 lies in one of the eight boxes specified in Theorem 1. There are no zeros on the coordinate axes.

Proof. At an axis point, one of the equations (13) would make a sum of two positive values zero. Thus any zero has positive coordinates. Choose any closed cell $J_i \times J_j$ containing a zero z. Equations (16) and (14) give

$$\|z - z_{ij}\|_\infty = \|M_{ij}^{-1} F_0(z)\|_\infty < K_{ij}\varepsilon. \tag{22}$$

If $\delta_{ij} > 0$, this would imply $\delta_{ij}/K_{ij} < \varepsilon$, contrary to (21), since $5 \cdot 10^{-12} < 1/5000$. Hence this is one of the eight cells in Table 1. Equations (20) and (22) now yield

$$\|z - z_{ij}\|_\infty < 256\varepsilon < 10^{-8} < r. \tag{23}$$

At the origin center the point is in $[0,r]^2$. The calculation includes all cells with an unbounded factor and all knot edges. It does not assume an a priori bound on the state. ∎

4 The eight analytic states

An exclusion certificate alone does not establish existence. We prove one actual analytic root in each box.

Lemma 3. *Every closed r-box centered at a nonzero point in Table 1 contains exactly one zero of F.*

Proof. Fix such a center $c = z_{ij}$ and put $M = M_{ij}$. The box avoids every coordinate knot by at least d. Equations (11) and (9) therefore imply, throughout the box,

$$\|DF(z) - M\|_\infty < 5\sigma. \tag{24}$$

For $N(z) = z - M^{-1}F(z)$, Lemma 2 gives

$$\|DN(z)\|_\infty < 1280\sigma =: q < \frac{1}{1000}, \qquad \|N(c) - c\|_\infty < 256\varepsilon. \tag{25}$$

The second bound uses $F_0(c) = 0$. Since $256\varepsilon + qr < r$, the map N is a contraction from the closed box into itself. Its unique fixed point is exactly a zero of F. The strict self-map estimate also places this zero in the box interior. ∎

Lemma 4. *The square $K = [0, r]^2$ contains exactly one zero of F, a diagonal point (ξ, ξ) with $0 < \xi < r$. Every orbit of T initialized in K converges to this point, and it is a local sink.*

Proof. For $0 \le t \le r$, the zero-knot term contributes at most $1/1000$ to each first derivative. All other knots are at distance at least $47/50 > d$. Hence

$$\begin{gathered} 0 < P'(t) < p_* := \frac{1}{1000} + 2\sigma, \qquad 0 < Q'(t) < q_* := \frac{1}{1000} + 3\sigma, \\ p_* + q_* < L := \frac{3}{1000}. \end{gathered} \tag{26}$$

Monotonicity and the value estimate give

$$0 < g(u, v) \le g(r, r) < \frac{2}{1000} r + \varepsilon < r \qquad ((u, v) \in K). \tag{27}$$

Thus $T(K) \subset K$. Put $w = P(v) + Q(u) \in [0, r]$. Direct differentiation gives

$$DT^2(u, v) = \begin{pmatrix} Q'(u) & P'(v) \\ P'(w)Q'(u) & Q'(v) + P'(w)P'(v) \end{pmatrix}. \tag{28}$$

Its first row sum is less than L, and its second is at most $q_* + p_* L < p_* + q_* < L$. Since K is convex, T^2 is an L-contraction of K. It has a unique fixed point z_*. The point Tz_* is also fixed by T^2 and belongs to K, so uniqueness gives $Tz_* = z_*$. Thus $z_* = (\xi, \xi)$. Strict positivity and (27) imply $0 < \xi < r$.

Even iterates converge by contraction, and odd iterates converge by continuity of T. The point is interior to K, and the same estimate bounds the spectral radius of $DT^2(z_*)$ by $L < 1$. It is therefore a hyperbolic local sink for T. ∎

The eight boxes are disjoint. The equations $F = 0$ are invariant under coordinate interchange. The unique zero in the upper diagonal box is therefore diagonal, while the six other zeros are exact swaps in paired boxes. None of those boxes meets the diagonal. Lemmas 3–4 and Proposition 3.1 prove exactly two equilibria and three prime two-cycles, with no additional states satisfying $T^2 z = z$ anywhere in the quadrant.

5 Return type at the actual states

At a state (u, v) with $T^2(u, v) = (u, v)$, write $p_u = P'(u)$, $q_u = Q'(u)$, and similarly for v. Since $T(u, v) = (v, u)$, the actual return derivative is

$$J = DT^2(u, v) = \begin{pmatrix} q_u & p_v \\ p_u q_u & q_v + p_u p_v \end{pmatrix}. \tag{29}$$

It is strictly positive, with determinant $q_u q_v > 0$. Its discriminant is

$$(q_u - q_v - p_u p_v)^2 + 4 p_u p_v q_u > 0. \tag{30}$$

Thus its two eigenvalues are distinct, positive and real. Put

$$\tau = \operatorname{tr} J = q_u + q_v + p_u p_v, \qquad \chi(1) = \det(I - J) = (1 - q_u)(1 - q_v) - p_u p_v. \tag{31}$$

For each nonzero affine center, evaluate these polynomials at the slopes of its cell and denote the results by $\tau_0, \chi_0(1)$. Exact substitution from (15) gives Table 2, and $\tau_0 - 2 > 1/2$ in every row. These additional finite rational checks are included in the companion certificate.

Center, or its swap	Bound on $\chi_0(1)$	Actual return type
$(1, 19/4)$	$\chi_0(1) < -1$	saddle
$(2, 9/2)$	$\chi_0(1) > 1/300$	repeller
$(3, 4)$	$\chi_0(1) < -1/20$	saddle
(ζ, ζ)	$\chi_0(1) > 1/150$	repeller

Table 2: Return types at the analytic roots in the indicated boxes. The first two columns are exact affine calculations; the proof transfers their strict signs to the analytic map.

At each actual root in these boxes, every scalar derivative differs from its cell slope by less than $e_* = 5\sigma$, by (24). All actual derivatives and comparison slopes are less than three. Expanding products in (31) therefore gives

$$|\tau - \tau_0| < 8e_*, \qquad |\det J - \det J_0| < 6e_*, \qquad |\chi(1) - \chi_0(1)| < 14e_*. \tag{32}$$

These rational errors preserve every sign in Table 2 and give $\tau > 2$. In particular the larger eigenvalue is greater than one. If $\chi(1) < 0$, the smaller eigenvalue lies in $(0, 1)$; if $\chi(1) > 0$, it is greater than one. This proves the stated types for all three actual cycles.

At the upper equilibrium, J is the square of $DT(\eta, \eta)$, and both eigenvalues of J exceed one. The eigenvalues of $DT(\eta, \eta)$ are real, of opposite signs, because its characteristic polynomial is $t^2 - P'(\eta)t - Q'(\eta)$. Their squares are the eigenvalues of J, so both have modulus greater than one. The lower equilibrium has already been treated separately in Lemma 4. It is positive and is a sink; it is not one of the seven expanding-return states.

6 The invariant basin boundary

We finish the proof of Theorem 1 by identifying the direction relevant to conditional stability. We use the global basin theorem and local boundary regularity of Berger–Duh [1, Theorem 2.2 and Remark 3.1] as prior results. Their source recurrence starts at indices $1, 2$; setting $y_n = x_{n-2}$ identifies it exactly with (1). Its topology is relative to I^2, and its basin partition uses the closure.

Lemma 1 verifies every assumption in (2). The source alternative in which every initial state has one common limit is impossible, since there are two distinct equilibria. Its other alternative applies. The basin supplied there contains $(0, 0)$, whose orbit converges to ξ by Lemma 4. It is therefore A_ξ, and the source gives its boundedness and convexity, (4), and convergence of both parity subsequences on its boundary.

The basin is relatively open. Indeed $U = [0, r)^2$ is a relatively open subset of the basin, and every orbit converging to (ξ, ξ) eventually enters U. Hence

$$A_\xi = \bigcup_{n \geq 0} T^{-n}(U). \tag{33}$$

Each nonlower periodic state is in neither A_ξ nor A_∞, so (4) places it on $B = \partial A_\xi$. The lower equilibrium is interior. Since $T^{-1}(A_\xi) = A_\xi$ and T is continuous, the boundary is forward invariant:

$$T(B) \subset B. \tag{34}$$

To see this, approximate a boundary point by basin points. Its image belongs to $\mathrm{cl}(A_\xi)$, and cannot belong to A_ξ by the preimage identity.

The basin is also coordinatewise downward closed. If $0 \leq z \leq w$ and $w \in A_\xi$, monotonicity gives

$$T^n(0, 0) \leq T^n z \leq T^n w. \tag{35}$$

Both bounds tend to (ξ, ξ), so $z \in A_\xi$. Consequently two boundary points cannot be strictly ordered in both coordinates: approximating the larger point by basin points would put the smaller point in the basin.

By the cited boundary regularity, B is a C^1 curve near each of the seven nonlower periodic states, all of which lie in the open quadrant. A boundary tangent cannot have two components of the same strict sign, since nearby points on such a curve would be strictly ordered. Invariance under T^2 makes its tangent line invariant under (29). A coordinate axis is not invariant under this strictly positive matrix. Its larger eigenvalue has a positive eigenvector, so the boundary tangent must be the other eigendirection:

$$T_z B = \ker(J - \lambda_- I). \tag{36}$$

This proves the direction assertion rather than inferring it from an ambient label.

In a local C^1 coordinate on B, the derivative of the two-step return at the periodic point is λ_-. If $0 < \lambda_- < 1$, continuity gives a smaller invariant interval on which the return is contracting. If $\lambda_- > 1$, it gives an interval on which distances from the fixed point expand until exit. Applying this to the signs in Table 2 proves conditional attraction for the first and third cycles and conditional repulsion for the middle cycle and upper equilibrium. For a two-cycle, convergence of the even iterates to one phase and continuity of T give convergence of the odd iterates to the other. The attracting cycles remain saddles in the ambient quadrant.

These statements include all admissible initial conditions in the source theorem. Axis states enter the open quadrant within two steps because $g > 0$. There are no higher-period starting states: any periodic orbit other than the lower equilibrium lies on B, where each parity subsequence converges. A convergent periodic subsequence is constant, so the prime period divides two. This is a consequence of the prior global theorem and the exact realization, not a new general restriction on monotone maps.

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